Aniruthan Rajendran

+1(929)-754-8993 | aniruth.255@gmail.com | linkedin.com | github.com | aneee004.github.io

Education

New York University (CGPA: 4.0/4.0) Sep 2025 – May 2027
Master of Science in Computer Science *Brooklyn, NY*
- **Relevant Coursework:** Foundations of Data Science, Big Data, Machine Learning, Algorithmic ML & DS

Indian Institute of Technology, Hyderabad (CGPA: 9.08/10) Aug 2017 – May 2021
B. Tech in Mechanical Engineering, with a Minor in Economics *Hyderabad, India*
- **Relevant Coursework:** Operating Systems, Database Management Systems, Algorithms

Experience

Senior Software Engineer Jul 2021 – May 2025
Apple Inc. *Hyderabad, India*
- Designed and built a large-scale, fault-tolerant log-data collection pipeline using inter-process communication and background services, ensuring correct, consistent, and concurrency-safe data capture under parallel execution and partial failures.
- Designed secure data handling pipelines using hybrid encryption schemes (AES + RSA), providing confidentiality guarantees.
- Designed reusable components for a picture-in-picture feature, handling OS-level windowing, focus, and fullscreen transitions.
- Developed synchronization infrastructure to coordinate execution across multiple devices using semaphore-based concurrency control, improving correctness under parallel workloads.
- Led end-to-end development across multiple projects: system design, POCs, milestone planning, code reviews, and repository ownership for solutions adopted across teams.

Software Engineering Intern Apr 2020 – Jun 2020
UST Global *Trivandrum, India*
- Developed a fault-tolerant streaming engine with Apache Kafka and Spark to process high-frequency video data in real time.
- Integrated ML models to detect face-mask and social distancing violations during the COVID-19 pandemic: D.V.A.S.P.

Projects

Big Data Analysis with Spark
- Designed and implemented an end-to-end distributed data pipeline in Spark over 10+ years of daily equity OHLCV data ($\sim$1M rows), using Parquet storage for efficient preprocessing and model training.
- Engineered time-series features and trained predictive models for next-day price movement.
- Modeled inter-asset relationships using rolling correlation graphs and trained a Graph Convolutional Network (GCN), achieving improved predictive performance over non-graph, factor-based baselines.

Heavy Light Decomposition (HLD)
- Implemented HLD for trees using **Segment Trees** & **Binary Lifting**, for **path queries** & **point updates** in $\mathcal{O}(\ln(n))$.
- Collaborated with moderators to improve code quality and add unit tests; accepted into TheAlgorithms/C-Plus-Plus.

Bayesian Inference on Crude Oil Data
- Designed a probabilistic modeling pipeline to analyze relationships between oil prices and short-term U.S. Treasury yields.
- Implemented a causal dependency graph to distinguish correlation from causation and control for confounding variables.
- Validated the model by identifying a consistent directional relationship between oil price movements and short-term yield changes, aligned with established economic literature.

Write Yourself a Git (W.Y.A.G)
- Implemented a minimal version-control system from first principles, including content-addressable storage, commits, branching, checkouts, and repository state management, to understand Git's internal data model and workflows.

Achievements

GRE	170 Quantitative, 163 Verbal	2023
ACM-ICPC	Rank 27	2020
National Math Olympiad	Cleared RMO and appeared for INMO	2015
Google KickStart, Round G 2020	Rank 150	2020

Technical Skills

Languages: Python, C/C++, Swift, Java, Bash
Libraries: pandas, NumPy, Matplotlib, Seaborn, PyMC, scikit-learn
Developer Tools: Git, Docker, Jenkins, AWS
Core Areas: Distributed Systems, Machine Learning, Graph ML, Statistical Modeling, Algorithms & Data Structures